Subnetting mastery & mock exam:

1. Given the network 192.168.10.0/24, divide it into 4 equal subnets and write down the network address, first usable IP, last usable IP, and broadcast address for each subnet.

ANS:

Divide 192.168.10.0/24 into 4 Equal Subnets

To create **4 equal subnets**, borrow **2 host bits**:

- Original: /24
- New prefix: **/26**
- Subnet mask: **255.255.255.192**
- Each subnet has **64 addresses**
- **62 usable IP addresses**

Subnet	Network Address	First Usable IP	Last Usable IP	Broadcast Address
1	192.168.10.0/26	192.168.10.1	192.168.10.62	192.168.10.63
2	192.168.10.64/26	192.168.10.65	192.168.10.126	192.168.10.127
3	192.168.10.128/26	192.168.10.129	192.168.10.190	192.168.10.191
4	192.168.10.192/26	192.168.10.193	192.168.10.254	192.168.10.255

Final Answer

4 subnets = /26

Each subnet provides **62 usable IP addresses**.

2. You are setting up a Zomato-style food delivery office with 3 departments: Customer Support (50 devices), Delivery Management (30 devices), and Marketing (20 devices). Design a subnetting plan using the 10.0.0.0/24 network so that each department has its own subnet, maximizing available addresses and minimizing waste. List the subnet mask and IP range for each department.

ANS:

VLSM Subnetting Plan

Base network: **10.0.0.0/24**

We use VLSM and allocate the largest department first.

Department	**Devices**	**Subnet**	**Subnet Mask**	**Usable IP Range**	**Usable IPs**
Customer Support	50	10.0.0.0/26	255.255.255.192	10.0.0.1 – 10.0.0.62	62
Delivery Management	30	10.0.0.64/27	255.255.255.224	10.0.0.65 – 10.0.0.94	30
Marketing	20	10.0.0.96/27	255.255.255.224	10.0.0.97 – 10.0.0.126	30

Why this is efficient?

- **Customer Support:** 50 devices → /26 gives 62 usable IPs.
- **Delivery Management:** 30 devices → /27 gives exactly 30 usable IPs.
- **Marketing:** 20 devices → /27 gives 30 usable IPs.
- No subnet overlaps.
- The remaining range **10.0.0.128 – 10.0.0.255** is available for future expansion.

Final subnet plan: /26 + /27 + /27 — efficient VLSM with minimal address waste.

**3. Troubleshoot the following scenario: Two teams in your Flipkart warehouse are accidentally assigned overlapping subnets (Team A: 172.16.10.0/25, Team B: 172.16.10.64/26). Identify the overlapping IP range and suggest a corrected subnet plan so both teams have unique IP ranges.

Hint: Draw the address ranges for both subnets to spot the overlap.**

ANS:

Troubleshooting Overlapping Subnets

Given

- **Team A:** 172.16.10.0/25
- **Team B:** 172.16.10.64/26

Step 1: Find the IP ranges

Team	Network	Usable IP Range	Broadcast
Team A	172.16.10.0/25	172.16.10.1 – 172.16.10.126	172.16.10.127
Team B	172.16.10.64/26	172.16.10.65 – 172.16.10.126	172.16.10.127

Overlapping Range

The overlap is:

172.16.10.64 – 172.16.10.127

For usable host addresses, the overlap is:

172.16.10.65 – 172.16.10.126

Corrected Subnet Plan

Give each team a separate, non-overlapping subnet:

Team	Corrected Subnet	Usable IP Range	Broadcast
Team A	172.16.10.0/25	172.16.10.1 – 172.16.10.126	172.16.10.127
Team B	172.16.10.128/26	172.16.10.129 – 172.16.10.190	172.16.10.191

Final Answer

The original subnets overlap because Team B's 172.16.10.64/26 lies inside Team A's 172.16.10.0/25 range.

A corrected plan is:

Team A → 172.16.10.0/25
Team B → 172.16.10.128/26

Now both teams have **unique, non-overlapping IP ranges**.

**4. You are managing a growing IPL Fantasy League app backend and notice that your 192.168.1.0/26 subnet is running out of IPs. Propose a solution to handle address exhaustion so new servers can be added without disrupting existing users.

Hint: Consider subnet resizing or migrating to a larger subnet.**

ANS:

Solution: Handle IP Address Exhaustion

The current subnet is:

192.168.1.0/26

- Total addresses: **64**
- Usable IPs: **62**
- Range: 192.168.1.1 – 192.168.1.62

Since the app is growing, the best approach is to **migrate to a larger subnet**, such as /25.

Proposed New Subnet

192.168.1.0/25

- Subnet mask: **255.255.255.128**
- Total addresses: **128**
- Usable IPs: **126**

- Usable range: 192.168.1.1 – 192.168.1.126
- Broadcast: 192.168.1.127

Migration Approach

1. Keep all existing servers using their current IPs (192.168.1.1–62).
2. Expand the subnet from **/26 to /25**.
3. New servers can use the additional range **192.168.1.63–126**.
4. Update the subnet mask on servers, DHCP, routers, and network configuration during a planned maintenance period.
5. Verify connectivity before adding new servers.

Final Answer

Migrate from 192.168.1.0/26 to 192.168.1.0/25.

5. After completing your subnetting speed round and mock exam, review your answers and identify 3 types of mistakes you made (e.g., miscalculating subnet masks, overlapping ranges, wrong broadcast address). For each, write how you would avoid the mistake next time.

ANS:

Review of Subnetting Mistakes

Mistake Type	**What I Did Wrong**	**How I Will Avoid It Next Time**
1. Wrong Subnet Mask	I sometimes selected the wrong / prefix for the required number of hosts.	I will first calculate **usable hosts = 2^n – 2** and then choose the smallest suitable subnet.
2. Overlapping Ranges	I accidentally assigned a subnet that was already inside another subnet's address range.	I will write the **network and broadcast address** of every subnet and check that the next subnet starts after the previous broadcast.

Mistake Type	What I Did Wrong	How I Will Avoid It Next Time
3. Wrong Broadcast Address	I sometimes calculated the broadcast address incorrectly.	I will use **Broadcast = next subnet's network address − 1** and verify the range before finalizing.